Taylor Daniels

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Location: Indianapolis, IN

About

Math PhD with expertise in asymptotic analysis and integer partitions, and significant background and experience in programming in and physics.

Languages: **English** – native; **Chinese (Mandarin)** – years of study, including teaching experience; **Spanish** – years of past experience and practice.

Some Work and Experience

Programming and Computation:

- **Large data sets for Doctoral Thesis:** Used multiple languages and interfaced with remote computation resources to compute long sequences of novel weighted integer-partitions; these explicit datasets informed and guided different theoretical results and works on said sequences. Implemented a mix of parallel and recursive algorithms to overcome (sub-)exponential search space.
  - Tools: **python**, **GP/Pari**, **remote computation servers**, **Sagemath**.
- **Domain coloring:** Produced "magnitude/phase" plots of complex-valued generating functions relevant to integer-partition sequences; used and modded open source library `cplot`.
  - Tools: **matplotlib**, **cplot**, **Mathematica**.
- **(Ongoing) Certificate in Data Science:** Participation in the Erdős Institute's Summer data science bootcamp/certificate program.
  - Tools: **python**, **pandas**, **machine learning**.

Graduate-level Math/Physics (Outside of PhD thesis work):

- **Gauge theory/fields:** Excess of one year of faculty-directed reading in gauge theory, including vector-bundle framework of fields, gauge symmetries, and differential geometry.
- **Quantum mechanics:** Full audit of graduate quantum mechanics course at Purdue; independent study of undergraduate material.
- **Lagrangian and Hamiltonian mechanics:** Undergraduate coursework.

Collaboration and Communication:

- **Coauthorship:** Experience collaborating on- and coauthoring math research manuscripts.
- **Course organization:** Lengthy experience in organizing and instructing college-level math courses, including Complex Analysis, Diff. Eq., and Lin. Alg.,. Students love me, apparently.

Education

2024	**Ph.D.**, Mathematics, Purdue Univ., West Lafayette IN.
	Advisors: Trevor Wooley and David (Ben) McReynolds
2015	**M.S.**, Mathematics, Univ. of Louisville, Louisville KY.
2014	**B.S.**, Mathematics, Univ. of Louisville, Louisville KY.

Employment

2024 – 2026	**Visiting Assistant Professor**, Purdue Univ. Mathematics
2015 – 2024	**Graduate Teaching/Research Assistant**, Purdue Univ. Mathematics
2015	**Instructor**, STARTALK Summer Program, Louisville KY
	An accelerated summer course in Mandarin Chinese, for students grade 6-8.
2012 – 2015	**Teaching Assistant**, Univ. of Louisville Mathematics

Awards

2021 – 2022	**Miscellaneous Scholarship**, Purdue Univ. Mathematics
Fall 2020	**Excellence in Teaching Award**, Purdue Univ. Mathematics
2009 – 2014	**Trustee's Scholarship**, Univ. of Louisville
2013 – 2014	**Bickel Scholarship**, Univ. of Louisville Mathematics
2012 – 2013	**Pedigo Scholarship**, Univ. of Louisville Mathematics
2009 – 2012	**Humana Scholarship**, Humana Inc., Louisville KY

Organizations

2014 – 2015	**Vice President**, Math Club, Univ. of Louisville
2012 – 2014	**Math Club**, Univ. of Louisville
2009 – 2012	**Chinese Learner's Society**, Univ. of Louisville
2010 – 2011	**President**, Chinese Learner's Society, Univ. of Louisville

Publications

T. Daniels, T. Huber, J. McLaughlin, and D. Ye, *The p-dissection of a product of quintuple products*, submitted; arXiv:2603.04666; 24 pp.

T. Daniels, *Vanishings of the Legendre-17 signed partition numbers*, submitted; arXiv:2603.02373; 49 pp.

T. Daniels, *Biasymptotics of the Möbius- and Liouville-signed partition numbers*, submitted; arXiv:2310.10617; 38 pp.

T. Daniels, *Vanishing coefficients in two q-series related to Legendre-signed partitions*, Res. Number Theory **10** (2024), Paper No. 81, 12 pp.. doi:10.1007/s40993-024-00566-x

T. Daniels, *The Asymptotics of Some Signed Partition Numbers*, Doctoral Thesis, August 2024. doi:10.25394/PGS.26364385.v1

T. Daniels, *Legendre-signed partition numbers*, J. Math. Anal. Appl. **542** (2025), no. 1., Paper No. 128717, 40 pp.. doi:10.1016/j.jmaa.2024.128717

T. Daniels, *Bounds on the Möbius-signed partition numbers*, Ramanujan J. **65** (2024), 81-123. doi:10.1007/s11139-024-00885-8

T. Daniels and D. Smith-Tone, *Differential Properties of the HFE Cryptosystem*, Post-Quantum cryptography, 2014, 59–75

Letters of Reference

Prof. Bruce Berndt, UIUC Mathematics, berndt at illinois dot edu
Prof. K. Joe Chen*, Purdue Univ., chenjk@purdue.edu
Dr. Dominic Naughton*, Purdue Univ., dnaughto at purdue dot edu
Prof. Trevor Wooley FRS (PhD co-advisor), Purdue Univ., twooley at purdue dot edu

*To provide letters concerning teaching.